

Supplementary Information for “From Aggregate
Accuracy to Branch Reliability: A Capability-First
Training Paradigm for Mixture-of-Experts
Physics-Informed Neural Networks”

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This Supplementary Information accompanies a manuscript submitted to *Computers & Mathematics with Applications*. It provides notation, metric definitions, complete training configurations, extended derivations, and per-seed results. Equation and theorem numbers explicitly identified as belonging to the main manuscript refer to that document.

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Appendix A. Notation

Table A.1: Main notation used in the paper.

Symbol	Meaning
$z = (x, y, t)$	space-time point
$\mathcal{Q}$	space-time domain $\Omega \times (0, T)$
u_b	shared base network output
e_k	residual correction of the k th expert
$u_k = u_b + \alpha e_k$	complete k th single-branch prediction
g_k, s_k	gating weight and logit
u_{mix}	gate-weighted mixture prediction
u^*	reference solution
δ_k	pointwise error of the k th branch
$E_{\text{mix}}, E_{\text{weighted}}$	mixture and weighted single-branch mean squared error
G_{agg}	soft aggregation gain
ℓ_k, k^*, Δ	local capability loss, oracle branch, capability gap
$\varepsilon_{\text{orc}}^2, \eta, \zeta$	oracle error, mean misrouting weight, non-oracle routing cost

Appendix B. Metric definitions

Let $\mathcal{E}$ denote the evaluation point set, u^* the reference solution, g_k the gating weights, u_k the complete single-branch predictions, and $\ell_k = |u_k - u^*|^2$. For Burgers, $\mathcal{E}$ is the 344,064-point subset sampled from the $513^2 \times 41$ conservative WENO5 reference.

- **Mixture relative L_2 :** $(\sum_{z \in \mathcal{E}} |u_{\text{mix}}(z) - u^*(z)|^2)^{1/2} / (\sum_{z \in \mathcal{E}} |u^*(z)|^2)^{1/2}$.
- **Single-branch relative L_2 :** the same with u_{mix} replaced by u_k .
- **Responsibility-region absolute RMSE:** $(|R_k|^{-1} \sum_{z \in R_k} |u_k(z) - u^*(z)|^2)^{1/2}$, with $R_k = \{z : g_k(z) \geq g_j(z), \forall j\}$. The mixture version replaces u_k by u_{mix} .
- **Oracle relative L_2 :** the mixture error with u_{mix} replaced by the pointwise best branch.
- **Soft-routing regret:** $|\mathcal{E}|^{-1} \sum_{z \in \mathcal{E}} (\sum_k g_k(z) \ell_k(z) - \min_k \ell_k(z))$.
- **Effective experts:** $\exp[-\sum_k \bar{g}_k \log(\bar{g}_k + 10^{-12})]$, where

$$\bar{g}_k = |\mathcal{E}|^{-1} \sum_{z \in \mathcal{E}} g_k(z).$$

- **Route entropy and minimum load:**

$$H_{\text{route}} = -|\mathcal{E}|^{-1} \sum_z \sum_k g_k(z) \log[g_k(z) + 10^{-12}], \quad \min_k \bar{g}_k.$$

- **Soft-aggregation gain:** $G_{\text{agg}} = 1 - E_{\text{mix}}/E_{\text{weighted}}$, where $E_{\text{mix}} = |\mathcal{E}|^{-1} \sum_z |u_{\text{mix}} - u^*|^2$ and $E_{\text{weighted}} = |\mathcal{E}|^{-1} \sum_z \sum_k g_k(z) \ell_k(z)$.
- **Top-1, temperature, and deletion ratios:** relative L_2 under argmax hard routing, the post-hoc transform $\tilde{g}_k(2) \propto g_k^{1/2}$, or deletion of the highest-load expert, divided by the original soft-mixture relative L_2 .

Appendix C. Training configuration

The main model is a residual soft MoE-PINN with 186,727 total parameters. The base network has input dimension 3 (x, y, t), output dimension 1, width 96, depth 5, and tanh activation. The four experts are smooth (width 48, depth 3), iso-shock (width 96, depth 5), directional-shock (width 96, depth 5), and wave (width 64, depth 4). The gate is a local-convolutional context gate of width 64 and depth 3 with learned-router temperature $T_g = 0.8$, 16 context neighbors, and a hidden convolution width of 24.

Primary training protocol: base pretraining for 1,000 steps (learning rate 10^{-3}); independent training of each expert for 1,000 steps (learning rate 10^{-3} , 1,800 supervised points per expert); static-teacher gate training for 1,000 steps (learning rate 5×10^{-4}); or co-adaptation as 1,000 joint updates with expert/gate learning rates $10^{-3}/5 \times 10^{-4}$. The teacher and test grids are spatially offset and have zero coordinate overlap. The lower-budget 2×2 screening control retains the 270/300-step configuration described in Section 4.2 of the main manuscript. Collocation points: $n_{\text{col}} = 6000$, $n_{\text{ic}} = 1200$, and 400 points per boundary face; loss weights $w_{\text{res}} = 1$, $w_{\text{ic}} = 5$, $w_{\text{bc}} = 2$, and $w_{\text{sup}} = 8$. Reference-teacher sharpness is $\beta = 8$; the capability-prior exponent is 1.15; the gate balance coefficient is 0.01; and targets are refreshed every ten co-adaptive steps. All experiments use Adam with default moments, Xavier initialization, gradient-norm clipping at one, deterministic seeds 42–51, and the same sampled sets within each paired seed. The recorded environment is Python 3.10.10, PyTorch 2.6.0+cu124, NumPy 2.2.6, and SciPy 1.15.3.

APINN configurations:

- parameter-matched two-subnetwork: shared and subnet widths 143/depth 3; gate width 48/depth 1; 186,429 parameters; scalar binary gate and official XPINN prior with prior temperature 0.32;

- official-size two-subnetwork: shared width 20/depth 2, subnet width 20/depth 3, gate width 20/depth 1; 3,583 parameters; the same XPINN prior;
- parameter-matched four-subnetwork: shared and subnet widths 112/depth 3, gate width 48/depth 2; 193,480 parameters; spatial prior.

The two-subnetwork models use 10,000 gate-pretraining steps; the four-subnetwork model uses 1,000. All use 8,000 joint steps, gate/train learning rates $10^{-3}/8 \times 10^{-4}$, weight decay 10^{-3} , $n_{\text{col}} = 6000$, $n_{\text{ic}} = 1200$, 400 boundary points per face, and residual/initial/boundary weights 1/10/5.

For Allen–Cahn, $u_t = \Delta u + \epsilon^{-2}(u - u^3)$ on $[-1, 1]^2 \times [0, 0.25]$, $\epsilon = 0.08$, $u(x, y, 0) = \tanh[(0.8 - \sqrt{x^2 + y^2})/(\sqrt{2}\epsilon)]$, and boundary value -1 . The reference uses exact reaction updates followed by Crank–Nicolson diffusion diagonalized by a discrete sine transform, with a 201^2 grid and $\Delta t = 2 \times 10^{-4}$; a 401^2 , 10^{-4} refinement gives the differences reported in Section 5.4. Three width-96/depth-5 experts represent interior, exterior, and interface behavior; the gate has width 48/depth 3 and $T_g = 0.8$. Each expert and the main phase use 1,500 steps.

For KdV, $u_t + 6uu_x + u_{xxx} = 0$ on $[-20, 20] \times [0, 6]$. The exact two-soliton solution is $u = 2\partial_{xx} \log F$, with $F = 1 + e^{\eta_1} + e^{\eta_2} + A_{12}e^{\eta_1 + \eta_2}$, $\eta_i = k_i x - k_i^3 t + \delta_i$, $(k_1, k_2) = (1, 0.5)$, $(\delta_1, \delta_2) = (0, 4)$, and $A_{12} = [(k_1 - k_2)/(k_1 + k_2)]^2$. The three experts have widths/depths 96/5, 64/3, and 96/5; the width-32/depth-3 gate uses $T_g = 0.9$. The full independent phase is 880 steps per expert, the gate-only phase 400 steps, and joint fine-tuning 2,200 steps; the schedule sweep is not update-budget matched.

Appendix D. Derivations and proof supplement

Appendix D.1. Proof of the nonlinear mixture-residual identity

Let $u = \sum_k g_k u_k$. Direct differentiation gives

$$u_t = \sum_k g_k u_{k,t} + \sum_k g_{k,t} u_k, \quad (\text{D.1})$$

$$Du = \sum_k g_k Du_k + \sum_k u_k Dg_k, \quad (\text{D.2})$$

$$\Delta u = \sum_k g_k \Delta u_k + 2 \sum_k \nabla g_k \cdot \nabla u_k + \sum_k u_k \Delta g_k. \quad (\text{D.3})$$

Substitution into $\mathcal{B}_\nu[u] = u_t + uDu - \nu\Delta u$, followed by adding and subtracting $\sum_k g_k u_k Du_k$, gives

$$\mathcal{B}_\nu[u] = \sum_k g_k (u_{k,t} + u_k Du_k - \nu\Delta u_k) \quad (\text{D.4})$$

$$+ \sum_k u_k (g_{k,t} - \nu\Delta g_k) - 2\nu \sum_k \nabla g_k \cdot \nabla u_k + u \sum_k u_k Dg_k \quad (\text{D.5})$$

$$+ u \sum_k g_k Du_k - \sum_k g_k u_k Du_k. \quad (\text{D.6})$$

The three lines are, respectively, the weighted branch residuals, the gate-derivative interaction, and the nonlinear interaction. Moreover,

$$\begin{aligned} u \sum_k g_k Du_k - \sum_k g_k u_k Du_k = & - \left[\sum_k g_k u_k Du_k \right. \\ & \left. - \left(\sum_k g_k u_k \right) \left(\sum_k g_k Du_k \right) \right], \end{aligned} \quad (\text{D.7})$$

which is $-\text{Cov}_g(u_k, Du_k)$ and proves the nonlinear mixture-residual identity in the main manuscript.

For two branches, the nonlinear term reduces to

$$\mathcal{I}_{\text{nl}} = -g_1 g_2 (u_1 - u_2)(Du_1 - Du_2). \quad (\text{D.8})$$

Thus disagreement in both branch value and directional derivative can generate a nonlinear interaction even for a constant gate.

Appendix D.2. Derivation of the gate-expert coupled gradient

By Eq. (1) of the main manuscript, the mixture output depends on the expert parameters θ_k through $u_{\text{mix}} = u_b + \alpha \sum_j g_j e_j$, so $\partial u_{\text{mix}} / \partial e_k = \alpha g_k$. The chain rule gives $\nabla_{\theta_k} \mathcal{J} = (D_{\theta_k} e_k)^* [\alpha g_k q]$. For the gate logit s_k , note that $\partial g_k / \partial s_k = g_k(1 - g_k)/T_g$ and $\partial g_j / \partial s_k = -g_j g_k / T_g$, hence

$$\frac{\partial u_{\text{mix}}}{\partial s_k} = \frac{\alpha}{T_g} \left[g_k(1 - g_k)e_k - \sum_{j \neq k} g_j g_k e_j \right] = \frac{\alpha}{T_g} g_k (e_k - \bar{e}). \quad (\text{D.9})$$

Substituting $D_{s_k} \mathcal{J}[\xi] = \langle q, \partial u_{\text{mix}} / \partial s_k \xi \rangle$ gives Eq. (2) of the main manuscript.

Appendix D.3. Expansion of the Jensen identity

Pointwise expansion of the mixture error:

$$\left| \sum_k g_k \delta_k \right|^2 = \sum_{i,j} g_i g_j \delta_i \delta_j = \sum_k g_k |\delta_k|^2 - \frac{1}{2} \sum_{i,j} g_i g_j |\delta_i - \delta_j|^2, \quad (\text{D.10})$$

where the second step uses $\sum_{i,j} g_i g_j |\delta_i - \delta_j|^2 = 2 \sum_k g_k |\delta_k|^2 - 2 \sum_{i,j} g_i g_j \delta_i \delta_j$. The cross term is nonnegative, so the mixture error does not exceed the weighted single-branch error.

Appendix D.4. Proof of the error bound

At a fixed point z , let k^* denote the oracle branch. Rewrite the mixture error as

$$u_{\text{mix}} - u^* = g_{k^*}(u_{k^*} - u^*) + \sum_{j \neq k^*} g_j(u_j - u^*). \quad (\text{D.11})$$

Pointwise Jensen gives $|\sum_k g_k(u_k - u^*)|^2 \leq \sum_k g_k |u_k - u^*|^2$. Separating the oracle term and using $g_{k^*} \leq 1$, then integrating over $\mathcal{Q}$, yields $\varepsilon_{\text{mix}}^2 \leq \varepsilon_{\text{orc}}^2 + \zeta$. Since $\zeta \leq M^2 \sum_{j \neq k^*} g_j = M^2 \eta$, the second inequality in Eq. (14) of the main manuscript follows.

Appendix E. Per-seed results

Table E.2: Per-seed main results on the conservative WENO5 reference solution.

Seed	Mode	Soft-mixture L_2	Worst-branch L_2	Top-1 ratio	$T=2$ ratio	Delete-expert ratio	Effective experts
42	CFST	0.2070	0.3270	1.091	0.962	1.192	3.339
42	coadapt	0.2377	32.39	1.427	2.378	1.341	2.619
43	CFST	0.1864	0.2753	1.059	0.979	1.007	3.713
43	coadapt	0.1326	10.28	1.259	2.326	2.210	2.731
44	CFST	0.2163	0.4092	1.031	1.019	1.221	3.591
44	coadapt	0.1751	24.01	1.419	3.267	1.571	2.814
45	CFST	0.2079	0.3237	1.088	0.967	1.056	3.419
45	coadapt	0.1362	29.81	1.208	5.156	1.411	2.732
46	CFST	0.1907	0.3347	1.042	1.001	1.074	3.611
46	coadapt	0.1328	13.99	1.340	2.065	1.173	2.505
47	CFST	0.2162	0.3276	1.040	1.001	1.008	3.716
47	coadapt	0.1553	34.41	1.395	2.715	1.135	3.000
48	CFST	0.2199	0.3149	1.035	0.992	1.058	3.815
48	coadapt	0.1509	19.89	1.604	3.786	2.209	2.535
49	CFST	0.1859	0.2779	1.066	0.977	1.022	3.833
49	coadapt	0.1217	19.81	1.348	3.257	1.440	2.601
50	CFST	0.2731	0.3880	1.047	0.985	1.037	3.551
50	coadapt	0.2129	15.65	1.454	2.067	1.385	2.535
51	CFST	0.2047	0.3576	1.080	0.990	1.225	2.917
51	coadapt	0.1444	14.86	1.377	1.972	1.320	2.563

Per-seed data show that co-adaptation attains a higher soft-mixture error only on seed 42, whereas its worst-branch error exceeds the CFST value on all ten seeds. The reliability contrast is therefore not seed-dependent, while the aggregate-accuracy advantage of co-adaptation is explicit rather than hidden by averaging.

Appendix E.1. APINN per-seed results

Table E.3: Per-seed APINN results on the conservative WENO5 reference (three seeds per version).

Version	Seed	Soft-mixture L_2	Worst-subnetwork L_2	Top-1/soft ratio
2-sub, parameter-matched	42	0.2644	0.968	2.223
2-sub, parameter-matched	43	0.2460	0.824	2.470
2-sub, parameter-matched	44	0.4508	0.850	1.262
2-sub, official size	42	0.2235	2.651	2.331
2-sub, official size	43	0.3281	1.880	3.340
2-sub, official size	44	0.1988	3.270	3.632
4-sub, parameter-matched	42	0.4403	1.576	1.022
4-sub, parameter-matched	43	0.3662	1.468	0.989
4-sub, parameter-matched	44	0.3638	1.319	1.037

Appendix E.2. Allen–Cahn per-seed results

Table E.4: Per-seed Allen–Cahn results (seeds 42–51). “Global branch” evaluates the complete interface branch over the full domain; “interface-region” restricts it to $|u^*| \leq 0.5$.

Seed	Mode	Mixture L_2	Global branch L_2	Interface-region L_2	Interface load
42	CFST	0.0362	1.0103	0.1818	0.0928
42	coadapt	0.2275	1.2846	1.9226	0.0220
43	CFST	0.0302	0.9221	0.1918	0.0897
43	coadapt	0.6549	1.6199	4.8195	0.0006
44	CFST	0.0260	1.1223	0.2028	0.1171
44	coadapt	0.1728	1.4371	4.5008	0.0000
45	CFST	0.0259	1.1141	0.2083	0.1013
45	coadapt	1.0352	1.5495	5.1028	0.0000
46	CFST	0.0398	1.0580	0.2191	0.1448
46	coadapt	0.6854	1.4684	4.8656	0.0004
47	CFST	0.0290	0.9945	0.2018	0.1434
47	coadapt	0.0400	0.6666	2.2650	0.0000
48	CFST	0.0490	1.1107	0.2028	0.0851
48	coadapt	0.2733	1.0056	1.0595	0.0000
49	CFST	0.0298	1.0639	0.1544	0.1064
49	coadapt	0.6859	1.3001	3.9545	0.0000
50	CFST	0.0353	1.0926	0.1809	0.0752
50	coadapt	0.8619	1.2259	3.5773	0.0000
51	CFST	0.0562	1.0847	0.4522	0.1450
51	coadapt	0.2361	1.5905	2.3834	0.0000